Experiment-9 Write the python to implement Travelling Salesman Problem

travelling.py - C:/Users/prana/OneDrive/Desktop/travelling.py (3.13.14)

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```
from itertools import permutations

a = [[0,10,15,20],
     [10,0,35,25],
     [15,35,0,30],
     [20,25,30,0]]

best = 9999

for p in permutations(range(1,4)):
    path = (0,) + p + (0,)
    cost = sum(a[path[i]][path[i+1]] for i in range(4))

    if cost < best:
        best = cost
        route = path

print("Shortest Route:", route)
print("Minimum Cost:", best)
```

IDLE Shell 3.13.14

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```
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
========== RESTART: C:/Users/prana/OneDrive/Desktop/travelling.py ==========
Shortest Route: (0, 1, 3, 2, 0)
Minimum Cost: 80
>>>
```